AI6124 / GROUP 4

Group Project Overview

Based on Week 8 lecture slides, “Neural in Neuro-Fuzzy Systems” (Project Briefing, pp. 16–40). A concise reference for our group discussion.

01 Project objective

Topic: Time Series Financial Price Prediction. Build a hybrid AI approach for predicting financial prices, compare it with a machine-learning benchmark, and examine whether predictions can improve simulated trading or portfolio performance.

Core project workflow

Stage	What we need to do
Financial dataset	Select historical price data for a stock, index, trust/REIT or similar financial instrument.
Hybrid AI model	Choose a course-related hybrid computational-intelligence method and explain how it fits the prediction task.
Benchmark	Compare with a relevant machine-learning / AI approach using appropriate, time-aware evaluation.
Trading evaluation	Study buy/sell signals and portfolio returns, with and without prediction.
Interpretation	Explain results, limitations, parameter effects and possible future work.

02 Hybrid AI is a key requirement

The proposed model must be a **hybrid AI method**, not just a standalone conventional neural network. Examples listed by the lecturer include:

- Neuro-fuzzy system (e.g., ANFIS)
- Neuro-evolution system
- Evolving neuro-fuzzy system
- Fuzzy RNN or Fuzzy LSTM

The slides note that readily available implementations (such as standard ANFIS code) may receive comparatively lower marks for method complexity than a recent hybrid method without readily available code or a meaningful extension of an established approach.

Required work and evaluation

03 Project rubrics (Week 8, p. 23)

Assessment area	Expected work
Dataset	Collect and describe the selected financial time series.
Hybrid AI method	Justify the model choice and its implementation.
Literature review	Trace relevant research chronologically; cite papers, applications and related methods.
Benchmark method	Select a comparator and appropriate evaluation design.
Implementation	Submit well-commented, reproducible Python code and results in a Jupyter Notebook.
Results analysis	Compare models and published work; discuss strengths, weaknesses and effects of parameters.
Future research	Describe realistic extensions and potential applications.
Presentation	Prepare concise, organized presentation slides covering the rubrics.

04 Evaluation expectations (pp. 34–36)

- **Preserve causality:** validation must respect temporal ordering and avoid future-data leakage.
- **Use appropriate metrics:** the slides explicitly mention RMSE and portfolio returns.
- **Go beyond numbers:** discuss why models perform differently and how parameters affect results.

05 Starter code and what remains (pp. 24–27)

Provided sample code	Student implementation
Download financial time-series prices; compute buy/sell signals; calculate profit/loss and portfolio performance.	Optimize trading rules and parameters; fit prediction models; evaluate prediction and trading performance on training and test data; explain the method and results.

Sample notebook: https://colab.research.google.com/drive/14VEWf5KzSxrTm8V4Px6cDu3eN3KUJXk5

Milestones and next steps

06 Project proposal: due 14 October 2026, 11:59 PM

Submit a **maximum of three PowerPoint slides** in bullet-point form:

Slide	Must cover
1. Dataset	One selected dataset, with a brief description (e.g., index, trust or company stock, period). Optional additional dataset if planned.
2. Hybrid AI	Named hybrid method plus a brief explanation of how it will be applied.
3. Benchmark	Selected benchmark machine-learning method.

One group member emails the proposal to all three lecturers and CCs the other member. Proposals can be amended later, but amendments require lecturer approval. Groups should differentiate their choice of dataset and/or method from other groups.

07 Key deadlines

Date (2026)	Deliverable
14 Oct, 11:59 PM	Proposal by email (one member, other member CC'd).
18 Nov, 11:59 PM	Final project: Jupyter Notebook and presentation slides; both students submit via NTULearn.
23 Nov, 11:59 PM	Receive projects / slides for peer review.
27 Nov, 11:59 PM	Peer-review scores submitted via email by one member.

Proposal submission recipients

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08 Decisions for our group

- Choose one financial instrument and historical period.
- Agree on a feasible hybrid AI architecture, supported by relevant literature.
- Select a benchmark, time-aware validation plan and evaluation metrics.
- Prepare and review the three-slide proposal before the deadline.

Optional bonus: Improve prediction or trading performance using extra inputs/datasets or transfer learning (examples mentioned include dividends and GDELT). This is not a requirement for the basic proposal.